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Based
Curriculum
(CBC)

PEAK ACHIEVERS[®]

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Raising Confident Learners for a Brighter Tomorrow

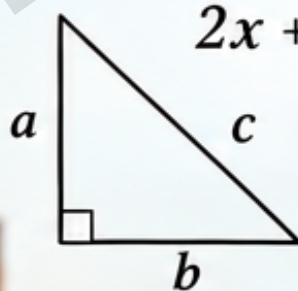
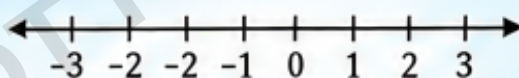
GRADE

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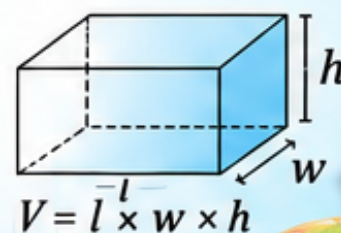
MATHEMATICS

BOOK 2

- ✓ Notes
- ✓ Examples
- ✓ Practice Exercises
- ✓ Real-Life Applications
- ✓ End-of-Strand Quizzes
- ✓ Mixed Concepts
- ✓ Five CBE Exam-Style Tests
- ✓ Answers & Marking Guides



$$2x + 3 = 11$$



NUMBERS

ALGEBRA

MEASUREMENTS

GEOMETRY

DATA HANDLING

Mathematics for Real Life, Real Skills, Real Opportunities

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Strand 4: Geometry - Angles; Geometrical Constructions

Strand 5: Data Handling and Probability - Data Handling

Mixed Concepts - 100 Questions

Five CBE Exam-Style Tests - 100 Marks Each

Answers and Marking Guides

STRAND 1: NUMBERS

1.1 Whole Numbers

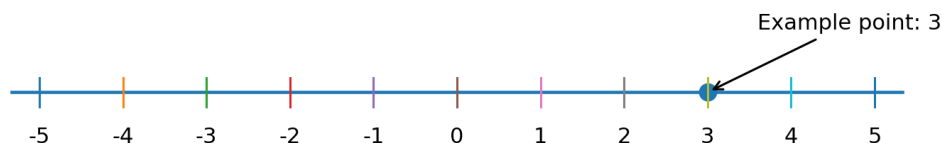


Figure 1. Number line for locating and comparing whole numbers.

Q: What are whole numbers? **A:** 0, 1, 2, 3, ...; they have no fractions or decimal parts.

Q: What is place value? **A:** The value of a digit according to its position in a number.

Q: How are whole numbers compared? **A:** Compare digits from the highest place value, moving from left to right.

Q: What is rounding? **A:** Replacing a number with a nearby convenient value at a stated place.

Q: What is BODMAS? **A:** Brackets, Orders, Division, Multiplication, Addition and Subtraction.

Practice

1. Write 607,305 in words.
2. State the place value of 8 in 4,836,219.
3. Round 487,649 to the nearest thousand.
4. Evaluate $96 - 48 \div 8 \times 5 + 14$.
5. Estimate $6,060 \div 30$.
6. A school has 2,475 Mathematics, 1,860 Science and 1,725 English books. Find the total.
7. The books in Question 6 are packed equally into 30 boxes. How many books are in each box?
8. If 24 boxes are issued, how many boxes remain and how many books are left?

Application Quiz

Community Tree-Planting Project: A = 3,485 trees, B = 4,206, C = 3,950, D = 4,359. Find the total, identify the highest and lowest, and calculate their difference. Round the total to the nearest thousand.

HINT: Align place values carefully before adding or subtracting.

1.2 Factors

Q: What is a factor? **A:** A whole number that divides another number exactly.

Q: What is a prime number? **A:** A number greater than 1 with exactly two factors: 1 and itself.

Q: What is HCF? **A:** The greatest factor common to two or more numbers.

Q: What is prime factorisation? **A:** Writing a number as a product of prime factors.

Practice

1. List all factors of 36.
2. State whether 29 is prime or composite.
3. Find the HCF of 48 and 72.
4. Prime-factorise 84.
5. Find the HCF of 84 and 126.
6. Use divisibility rules to test 4,590 for divisibility by 2, 3, 5, 9 and 10.

Application: 48 avocado seedlings and 72 mango seedlings are arranged into the greatest possible number of identical groups. Find the number of groups and the number of each type per group.

1.3 Fractions

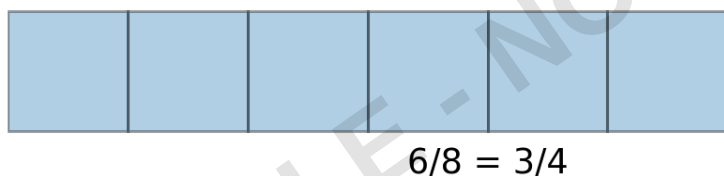


Figure 2. Fraction model showing $6/8 = 3/4$.

Q: What is a proper fraction? **A:** A fraction whose numerator is less than its denominator.

Q: How is a fraction simplified? **A:** Divide numerator and denominator by their HCF.

Q: How are unlike fractions added? **A:** Express them using a common denominator, then add the numerators.

Q: How are fractions multiplied? **A:** Multiply numerators together and denominators together, then simplify.

Q: How are fractions divided? **A:** Multiply the first fraction by the reciprocal of the second.

Worked example: $3/4 \times 2/5 = 6/20 = 3/10$.

Practice

1. Simplify $36/48$.
2. Convert $3 \frac{2}{5}$ to an improper fraction.
3. Evaluate $3/4 + 5/8$.
4. Evaluate $5/6 - 3/8$.

5. Find $\frac{3}{5}$ of 450.

6. Evaluate $\frac{5}{6} \div \frac{2}{3}$.

Application: A 1,200 L tank has $\frac{1}{4}$ of its water used, then $\frac{1}{3}$ of the remaining water is used. Find the water left.

 **THINK BIG:** From 960 kg of maize, $\frac{3}{8}$ is sold on Monday and $\frac{2}{5}$ of the remainder on Tuesday. Find the mass left and the mass per bag if the remainder fills 24 equal bags.

1.4 Decimals

Q: What is a decimal? **A:** A number written using a decimal point to show whole and fractional parts.

Q: How are decimals added or subtracted? **A:** Align decimal points and corresponding place values.

Q: What happens when multiplying by 10, 100 or 1000? **A:** Digits shift left by 1, 2 or 3 place-value positions respectively.

Q: How is a decimal rounded? **A:** Check the digit immediately to the right of the required place.

Practice

1. Evaluate $24.75 + 8.96 - 5.40$.

2. Calculate $36.8 \div 10 + 4.25$.

3. Round 18.746 to two decimal places.

4. Write $\frac{3}{4}$ as a decimal.

5. Arrange 2.45, 2.8, 2.35 and 2.75 in ascending order.

Application: 12 books cost KSh 175.50 each, 8 rulers cost KSh 24.25 each and 15 pens cost KSh 32.75 each. Find the total cost and balance from KSh 3,000.

1.5 Squares and Square Roots

n	1	2	3	4	5	6
n^2	1	4	9	16	25	36
n	7	8	9	10	11	12
n^2	49	64	81	100	121	144

Figure 3. Perfect squares from 1^2 to 12^2 .

Q: What is the square of a number? **A:** The product obtained when the number is multiplied by itself.

Q: What is a square root? **A:** A number that gives the original number when multiplied by itself.

Q: What is a perfect square? **A:** A number that is the square of a whole number.

Perfect squares 1-12: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

Practice

- Find 15^2 .
- Find $\sqrt{225}$.
- Find $\sqrt{324}$.
- A square courtyard has area 400 m^2 . Find its side and perimeter.
- A square assembly area has area 324 m^2 . Find its side and perimeter.

End-of-Strand 1 Quiz - Numbers

- Write 5,083,406 in words.
- Find the HCF of 84 and 126.
- Simplify $45/60$.
- Evaluate $7/8 - 5/12$.
- Find $4/9$ of 720.
- Evaluate $54.80 - 17.65 + 8.25$.

7. Find $\sqrt{361}$.
8. A school has 720 learners. $\frac{3}{8}$ are in Grade 7. Find the number.
9. 48 boxes contain 36 books each. Find the total.
10. A square courtyard has area 400 m^2 . Find its perimeter.

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